
Old Exam Questions AI in Life Sciences

Worked Solutions

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How this file is organized

This is the **solutions companion** to `exams.tex`. Every question is followed *immediately* by a green **answer box**. Inside it, each option is marked ✓ (correct) or ✗ (incorrect), with a short justification — *why* the correct options are correct and, for the tricky / near-miss distractors, *why* they are wrong. Questions come from two sittings: the **2024** exam (25 June 2024, Sections 1–6, *multiple-select* — one *or more* options correct) and the **2026** exam (25 June 2026, Sections 7–10, *single-best-answer* — exactly one option correct). Wording has been paraphrased; the answer options are kept intact. For a clean practice set with blank answers, use `exams.tex`.

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Drug Discovery, Molecules & Representations

Question 1: SMILES strings

Regarding the SMILES notation for molecules, which of the following statements hold?

- ☐ SMILES stands for *Simplified Molecular Input Line Entry System*.
- ☐ SMILES strings represent molecules in a linear sequence of characters.
- ☐ SMILES cannot be converted into a molecular graph.
- ☐ SMILES is always unique for each molecule, ensuring no two different molecules share the same SMILES.

Answers

- ✓ **A.** It is exactly the *Simplified Molecular Input Line Entry System*.
- ✓ **B.** A molecule is encoded as a 1D string of characters (atoms, bonds, ring/branch tokens).
- ✗ **C.** SMILES *can* be parsed into a molecular graph — that round-trip is one of its main uses.
- ✗ **D.** Not unique: a single molecule has *many* valid SMILES. Only a *canonical* SMILES (a fixed algorithm) gives one representative string — the plain format is not inherently unique.

Question 2: Reading a SMILES string

Consider the molecule *pyruvic acid* (2-oxopropanoic acid), i.e. a methyl group followed by a ketone carbonyl and then a carboxylic acid ($\text{CH}_3\text{-C(=O)-C(=O)-OH}$). Which of the SMILES strings below correctly encode this molecule?

- ☐ O=C(C(=O)O)C
- ☐ CC(=O)C(=O)O
- ☐ CCC000
- ☐ OCCOO

Answers

- ✓ **A.** O=C(C(=O)O)C: a carbonyl carbon bearing a C(=O)O (carboxyl) and a C (methyl) — exactly $\text{CH}_3\text{-CO-COOH}$, just written starting from the ketone.
- ✓ **B.** CC(=O)C(=O)O: methyl, ketone carbonyl, carboxyl read left-to-right — the same molecule.
- ✗ **C.** CCC000 is a chain of three carbons then three oxygens (no double bonds, no carboxyl) — not pyruvic acid, and chemically nonsensical.
- ✗ **D.** OCCOO encodes a short O/C chain with the wrong connectivity and no ketone/carboxyl — not the target.

Question 3: Molecular fingerprints

Which of the following statements about molecular fingerprints are accurate?

- ☐ A unique molecular graph can be constructed from them.
- ☐ They are binary or numerical representations of molecular structures.
- ☐ They are used for similarity searching and machine learning tasks.
- ☐ They provide a 3D representation of molecules.

Answers

- ✗ **A.** They are *lossy* and often hashed (bit collisions), so you cannot reconstruct a *unique* molecular graph from them.
- ✓ **B.** Fingerprints are fixed-length *binary* (or count/numerical) vectors encoding the presence of substructures.
- ✓ **C.** Their classic uses are Tanimoto *similarity search* and as ML features.
- ✗ **D.** They encode 2D substructure/topology, *not* 3D geometry.

Question 4: Bioassays

Select the correct statements about bioassays.

- ☐ They can be used to measure the biological response or function elicited by a substance.
- ☐ They typically test the substance at various concentrations.
- ☐ They are wet-lab procedures to acquire knowledge of a compound in a biological context.
- ☐ They are typically tested *in vitro*, i.e. in a living organism.

Answers

- ✓ **A.** The whole point of an assay is to quantify a biological response/function.
- ✓ **B.** They are run over a concentration range (dose–response), giving e.g. IC₅₀.
- ✓ **C.** They are physical wet-lab experiments that generate the bioactivity labels we later model.
- ✗ **D.** The trap is the definition: *in vitro* means “in glass” (test tube / cell culture), *not* in a living organism. “In a living organism” is *in vivo*.

Question 5: ADME properties

Which statements about ADME (Absorption, Distribution, Metabolism, and Excretion) properties are correct?

- ☐ Lipinski’s rule of five can be used as a proxy for ADME properties.
- ☐ Good ADME properties guarantee that a compound will be an effective drug.
- ☐ ADME properties such as clearance are often difficult to model.
- ☐ ADME properties are critical in determining the pharmacokinetics of a drug.

Answers

- ✓ **A.** Lipinski's Rule of Five is a cheap drug-likeness heuristic used as an ADME/oral-bioavailability proxy.
- ✗ **B.** Necessary but *not sufficient*: good ADME means the drug reaches its target intact, but efficacy still needs actual target binding and the right pharmacodynamics.
- ✓ **C.** Properties like clearance depend on complex metabolism and are notoriously hard to predict.
- ✓ **D.** ADME *is* pharmacokinetics — what the body does to the drug over time.

Question 6: Characteristics of QSAR / bioactivity data

Which of the following correctly characterise typical QSAR / bioactivity datasets?

- ☐ QSAR datasets usually have data points in the order of thousands or millions.
- ☐ QSAR datasets typically have a balanced distribution of active and inactive compounds.
- ☐ QSAR datasets often contain series of structurally related compounds.
- ☐ QSAR data often includes noise due to biological variability and experimental inconsistencies.

Answers

- ✗ **A.** Bioactivity data is usually *small/limited* (hundreds to a few thousand per target), not millions.
- ✗ **B.** It is typically strongly *imbalanced* — far more inactives than actives.
- ✓ **C.** Medicinal-chemistry projects explore *analog series* (one scaffold, small variations), so related compounds cluster heavily.
- ✓ **D.** Assays vary between labs/days and have measurement error, so labels are noisy.

Question 7: Applications of QSAR modeling

For which of the following applications is QSAR modeling relevant?

- ☐ Creating 3D protein structure models.
- ☐ Identifying potential drug candidates.
- ☐ Predicting the toxicity of new chemical compounds.
- ☐ Mapping the genome sequences of organisms.

Answers

- ✗ **A.** 3D protein structure is *structure prediction* (e.g. AlphaFold), a different problem — QSAR maps molecule → property.
- ✓ **B.** QSAR ranks/scores compounds by predicted activity — i.e. finds candidates.
- ✓ **C.** Predicting properties like toxicity (QSPR) is a core QSAR application.
- ✗ **D.** Genome sequencing/mapping is genomics, unrelated to QSAR.

Virtual Screening & QSAR Modeling

Question 8: Ligand-based virtual screening

Which statements about ligand-based virtual screening are correct?

- ☐ Molecular fingerprints are commonly used in ligand-based virtual screening.
- ☐ Ligand-based virtual screening is not suitable for predicting the bioactivity of unknown compounds.
- ☐ Ligand-based virtual screening requires a 3D structure of the target protein.
- ☐ Ligand-based virtual screening typically only considers bioactivity predictions to select compounds.

Answers

- ✓ **A.** Fingerprint similarity to known actives is the workhorse of ligand-based VS.
- ✗ **B.** It is precisely *used* to predict activity of unknown/untested compounds by analogy to known actives.
- ✗ **C.** Needing the protein's 3D structure is *structure-based* VS. "Ligand-based" means you only use known ligands, no target structure.
- ✗ **D.** Selection also weighs ADME, drug-likeness and structural-alert filters — not bioactivity alone.

Question 9: Data splits in QSAR model evaluation

When evaluating QSAR models, which of the following statements about data-splitting strategies hold?

- ☐ Temporal splits help evaluate the model's performance on future, unseen data by simulating real-world data acquisition over time.
- ☐ Generally, a random split is the best method for evaluating QSAR models as it eliminates biases.
- ☐ Cross-validation is rarely used in the evaluation of QSAR due to computational cost.
- ☐ Cluster splits group similar compounds together, ensuring that structurally similar compounds are not both in the training and test sets.

Answers

- ✓ **A.** Splitting by time mimics how data really arrives and tests genuine prospective generalisation.
- ✗ **B.** A *random* split is the classic trap: structurally similar compounds leak into both sets (compound-series bias), so it *over-estimates* performance rather than eliminating bias.
- ✗ **C.** Cross-validation *is* routinely used in QSAR; cost is not a reason to avoid it here.
- ✓ **D.** Scaffold/cluster splits keep an entire analog series on one side, so the test set is not just memorised neighbours.

Question 10: Early-recognition metrics in virtual screening

Because only the top of a virtual-screening ranking is followed up experimentally, recognising actives early matters most. Which of the following metrics are specifically designed to reward strong early recognition?

- ☐ Precision-Recall Curve.
- ☐ BEDROC (Boltzmann-Enhanced Discrimination of ROC).
- ☐ Average Precision (AP).
- ☐ Enrichment Factor (EF).

Answers

- ✗ **A.** A Precision–Recall curve summarises the *whole* ranking, not specifically the top.
- ✓ **B.** BEDROC applies an exponential weight that emphasises actives found *early* in the ranking.
- ✗ **C.** Average Precision is a general ranking metric and is not early-recognition-weighted.
- ✓ **D.** The Enrichment Factor measures how many actives appear in the top $x\%$ — a pure early-recognition metric.

Question 11: Target identification / deconvolution

Which statements about target identification / deconvolution are correct?

- ☐ For a given molecule, it consists of identifying to which protein it will most likely bind if multiple are present simultaneously.
- ☐ Multi-class classification can be used for it.
- ☐ Multi-task classification can be used for it.
- ☐ For a given molecule, it consists of identifying all the proteins it will bind if multiple are present simultaneously.

Answers

- ✓ **A.** Deconvolution asks: of the candidate proteins, which is the *most likely* target of this molecule.
- ✓ **B.** Picking one target out of several is naturally a *multi-class* (one-of- K) problem.
- ✗ **C.** *Multi-task* classification predicts activity against *many* targets independently — that is target *prediction*, not deconvolution.
- ✗ **D.** Identifying *all* binding proteins is again target prediction (many yes/no labels), not the “single most likely” framing of deconvolution.

Question 12: Drug synergy and antagonism

Which of the following statements about drug synergy and antagonism are correct?

- ☐ Drug synergy occurs when the combined effect of two or more drugs is greater than the sum of their individual effects.
- ☐ The Loewe additivity model is commonly used to assess drug synergy.

- ☐ Antagonistic effects occur when the combined effect of two or more drugs is greater than the sum of their individual effects.
- ☐ Drug synergy / antagonism evaluation is only applicable to chemically similar drugs.

Answers

- ✓ **A.** Synergy = the combination does *more* than the sum of the parts (super-additive).
- ✓ **B.** Loewe additivity is the standard reference model for “no interaction”, against which synergy is measured.
- ✗ **C.** This is the inverted definition: *antagonism* is when the combined effect is *less* than the sum. The sentence describes synergy but labels it antagonism.
- ✗ **D.** Combinations are studied for *any* drug pair (often deliberately different mechanisms) — chemical similarity is irrelevant.

Question 13: Random Forests vs. Graph Neural Networks

Comparing Random Forests (RF) and Graph Neural Networks (GNNs) for molecular modeling, which statements are correct?

- ☐ GNNs are inherently better for QSAR modeling because they naturally capture the graph-like structure of molecules, leading to more accurate predictions in all cases.
- ☐ RF models are less prone to overfitting compared to GNNs, making them a more reliable choice for datasets with a large number of features but limited samples.
- ☐ RF models are typically more interpretable than GNNs, making it easier to understand the contribution of individual molecular features to the model's predictions.
- ☐ GNNs are less sensitive to the quality and variability of the input data compared to RFs, making them more consistent in performance across different datasets.

Answers

- ✗ **A.** “Better in *all* cases” is the giveaway — on limited/noisy data RF often matches or beats GNNs (no free lunch).
- ✓ **B.** On small, high-dimensional QSAR datasets RF (bagging) is robust and overfits less than a data-hungry GNN.
- ✓ **C.** RF gives feature-importance scores over interpretable descriptors; GNN representations are opaque.
- ✗ **D.** The opposite is true: GNNs are typically *more* sensitive to data quantity/quality, not less.

Graph Neural Networks & Advanced Methods

Question 14: Molecular graphs and graph neural networks

Which statements about molecular graphs and graph neural networks are correct?

- ☐ A molecular graph can be represented by two matrices: one storing the node features and another storing the edges between the nodes.
- ☐ Node features can include atomic number, hybridization state, and formal charge.
- ☐ In a GNN only the node features are used to make the final predictions.
- ☐ The information about the bond type (e.g. single bond, double bond) cannot be included in a GNN.

Answers

- ✓ **A.** Standard encoding: a node-feature matrix plus an adjacency/edge matrix.
- ✓ **B.** Typical atom features are exactly atomic number, hybridisation, formal charge (one-hot/numeric).
- ✗ **C.** Message passing also uses *edge* information and the graph topology, not node features alone.
- ✗ **D.** Bond type *can* be encoded as edge features — single/double/aromatic bonds are routinely included.

Question 15: Limitations of GNNs in molecular modeling

Which of the following are genuine limitations of GNNs in the context of molecular modeling?

- ☐ GNNs are incapable of predicting any molecular properties accurately.
- ☐ GNNs can have difficulty capturing long-range interactions between atoms.
- ☐ GNNs may struggle with very large and complex molecular graphs.
- ☐ GNNs require a significant amount of computational resources for training.

Answers

- ✗ **A.** Far too strong — GNNs predict many molecular properties very well; the statement is an obvious overstatement.
- ✓ **B.** A fixed number of message-passing hops limits how far information travels (over-smoothing / under-reaching).
- ✓ **C.** Large, complex graphs strain both expressivity and memory.
- ✓ **D.** Training GNNs is compute- and memory-intensive relative to e.g. RF.

Question 16: Multi-modal contrastive learning

Which statements about multi-modal contrastive learning are correct?

- ☐ Multi-modal contrastive learning encodes multiple modalities (e.g. molecules and proteins) into a shared latent space.
- ☐ Multi-modal contrastive learning is a self-supervised learning technique.

- ☐ Multi-modal contrastive learning concentrates (latent) representations from multiple modalities (e.g. molecules and proteins) to make predictions.
- ☐ Multi-modal contrastive learning is an unsupervised learning technique.

Answers

- ✓ **A.** The defining idea: map both modalities into one *shared* embedding space (à la CLIP).
- ✓ **B.** It is *self-supervised* — supervision comes from matched/mismatched pairs, not human labels.
- ✗ **C.** It *aligns* matched pairs (pull together / push apart); “concentrates representations to make predictions” mischaracterises the objective.
- ✗ **D.** The subtle distinction: it is self-supervised, *not* unsupervised — the pairing provides the training signal.

Question 17: Multiple instance learning (MIL)

Which of the following statements about multiple instance learning (MIL) hold?

- ☐ MIL involves training on sets or bags of instances where each instance is individually labelled.
- ☐ MIL requires the set/bag size to be fixed.
- ☐ MIL involves training on sets or bags of instances that have a single label.
- ☐ MIL models should be invariant to the permutation of instances.

Answers

- ✗ **A.** Exactly what MIL is *not*: instances are *not* individually labelled (that would be ordinary supervised learning).
- ✗ **B.** Bags can have varying sizes; the aggregation (e.g. attention pooling) handles any number of instances.
- ✓ **C.** The hallmark of MIL: a label is attached to the *bag*, not to each instance.
- ✓ **D.** A bag is a set, so the model output must be *permutation-invariant*.

Generative Models & Chemical Synthesis

Question 18: Generative models for small molecules

Which statements about generative models for small molecules are correct?

- ☐ As the syntax of SMILES strings is complicated and with long dependencies, generators of SMILES representations will produce less than 10% valid SMILES strings.
- ☐ Generative models are used to explore the known chemical space of synthesized molecules.
- ☐ Generative adversarial networks cannot be used to generate small molecules.
- ☐ Autoregressive models can be used to generate SMILES representations of small molecules.

Answers

- ✗ **A.** Empirically false: an LSTM reaches ~98% valid SMILES, and SELFIES gives 100% by construction — nowhere near “<10%”.
- ✗ **B.** Generative models explore *novel* chemical space, not just the already-synthesised *known* set.
- ✗ **C.** GANs *can* generate molecules (as can VAEs, normalizing flows, diffusion).
- ✓ **D.** Autoregressive (next-token) models generate SMILES one character at a time and work well.

Question 19: Goal-directed molecule generation

Which of the following correctly describe goal-directed molecule generation?

- ☐ Goal-directed generation focuses only on replicating existing molecular structures with desired properties.
- ☐ Goal-directed generation aims to create molecules with similar properties to the training set.
- ☐ Reinforcement learning can be used to optimize goal-directed molecule generation.
- ☐ Goal-directed generation aims to create molecules with specific desired properties.

Answers

- ✗ **A.** “Replicating existing structures” describes *distribution learning*, not goal-directed design.
- ✗ **B.** “Similar to the training set” is again *distribution learning* — goal-directed deliberately pushes *beyond* the training distribution.
- ✓ **C.** A scoring function + RL loop (generate, score, reward, update) is a standard goal-directed recipe (e.g. REINVENT).
- ✓ **D.** The aim is molecules with *specific desired* properties, optimised toward an objective.

Question 20: Fréchet ChemNet Distance (FCD)

Which statements about the Fréchet ChemNet Distance (FCD) and how it is computed are correct?

- ☐ FCD measures the distance between individual atoms of molecules.
- ☐ FCD is calculated by comparing the means and covariances of the latent representations of molecules.
- ☐ FCD is calculated directly on the raw molecular structures without any transformations.
- ☐ FCD is a metric used to compare the distribution of generated molecules to that of real molecules.

Answers

- ✗ **A.** It works on whole-molecule embeddings, not distances between individual atoms.
- ✓ **B.** FCD fits Gaussians to ChemNet’s last-hidden-layer embeddings and compares their *means and covariances*.

- ✗ C. It is computed in a *learned latent space* (ChemNet), not on raw structures — there is a transformation step.
- ✓ D. It is a *distributional* metric: how close is the generated set to the real set overall.

Question 21: AI for chemical reactions

Which of the following statements about machine learning for chemical reactions are correct?

- ☐ Some methods rely on so-called reaction templates, which are transformation rules for molecules.
- ☐ Deep learning methods that predict outcomes of chemical reactions can be trained on databases containing known chemical reactions.
- ☐ Transformer architectures have been used to predict outcomes of chemical reactions.
- ☐ Predicting chemical reactions with machine learning requires data from quantum mechanical calculations due to bond changes and electron movements.

Answers

- ✓ A. Template-based (neural-symbolic) methods select a reaction template = a transformation rule.
- ✓ B. Models are trained on reaction databases like USPTO (mined from patents).
- ✓ C. The “Molecular Transformer” treats reaction prediction as seq-to-seq translation.
- ✗ D. No QM data needed — predictions come from *reaction databases*, not quantum-mechanical calculations.

Question 22: Synthesis planning

Which statements about synthesis planning are correct?

- ☐ Synthesis planning involves designing a sequence of chemical reactions to produce a target compound.
- ☐ Retrosynthetic analysis is a key technique used in synthesis planning to break down complex molecules into simpler starting materials.
- ☐ Synthesis planning has been made trivial by modern deep learning methods.
- ☐ Synthesis planning involves only selecting the reactants.

Answers

- ✓ A. It designs a whole *sequence* of reactions from purchasable materials to the target.
- ✓ B. Retrosynthesis recursively breaks the target into simpler precursors — the core technique.
- ✗ C. Far from trivial: ~50% solve rates, and whether routes actually run under real conditions remains uncertain.
- ✗ D. It is much more than “selecting reactants” — it is multi-step route design with scoring and search.

Question 23: Retrosynthesis as a reinforcement-learning game

Multi-step retrosynthesis can be cast as a game solvable by reinforcement learning. Which of the following statements about this “retrosynthesis game” are correct?

- ☐ The actions in the retrosynthesis “game” involve selecting reactants for the reactions.
- ☐ The state in the retrosynthesis “game” refers to the set of available building blocks.
- ☐ The environment in the retrosynthesis “game” performs the reaction and returns new molecules.
- ☐ The reward in the retrosynthesis “game” is obtained when the product is successfully synthesized from the building blocks.

Answers

- ✓ **A.** An *action* is choosing a disconnection / set of reactants (a reverse reaction).
- ✗ **B.** The classic trap: the *state* is the current set of molecules *still to be made*, not the set of available building blocks (those are the goal you are trying to reach).
- ✓ **C.** The *environment* applies the chosen reaction and returns the new set of molecules.
- ✓ **D.** The *reward* comes when every remaining molecule is an available building block (the game is won).

Medical Data & Imaging

Question 24: Tabular medical data: types and missing values

Suppose we are given a table of patient medical records with three columns: “smoking status” (one of three string values: non-smoker, former smoker, current smoker), “cholesterol level” (floating-point measurements in mg/dL), and “diabetes risk score” (integer values in [0, 10] indicating the likelihood of developing diabetes). Which of the following statements are correct?

- ☐ Missing values in “cholesterol level” can be replaced by the column’s median value to handle skewness.
- ☐ Column “smoking status” is continuous, as it captures the various smoking habits of patients.
- ☐ “cholesterol level” is ordinal since cholesterol measurements can be arranged in increasing order.
- ☐ Missing values in “smoker status” can be replaced by the column’s mean value to handle skewness.

Answers

- ✓ **A.** Cholesterol is numeric; the *median* is the robust choice for imputing a skewed continuous column.
- ✗ **B.** Smoking status is *categorical* (three discrete labels), not continuous.
- ✗ **C.** Being orderable is not enough: cholesterol is a true *continuous* (interval/ratio) measurement with meaningful differences, not merely ordinal.

- ✗ **D.** You cannot take a *mean* of a categorical column — a missing smoking status would be imputed with the *mode*, not the mean.

Question 25: Batch effects in medical data

Which statements about batch effects in medical data are correct?

- ☐ Ignoring batch effects can lead to models that distinguish between batches rather than the actual biological conditions.
- ☐ Batch effects can be eliminated by simply increasing the sample size.
- ☐ Batch effects can create artificial differences in data due to variations in experimental conditions.
- ☐ Batch effects are usually beneficial as they introduce necessary variability into the data.

Answers

- ✓ **A.** If batches correlate with the label, the model takes the shortcut and separates *batches* instead of biology.
- ✗ **B.** More samples from the *same* biased batches does not remove the systematic offset — you need design/correction (randomisation, ComBat, etc.).
- ✓ **C.** They are exactly artificial, non-biological differences from instruments/reagents/handling.
- ✗ **D.** They are a confounder, not a benefit — this is the opposite of correct.

Question 26: Domain shifts in medical and healthcare data

Medical and healthcare data are frequently subject to domain shifts. Using the lecture notation (y = disease status, e.g. “healthy”/“diseased”; x = patient features, e.g. age, blood parameters, heart rate), which of the following statements about domain shifts are correct?

- ☐ If a machine learning model has a high predictive quality on a randomly drawn validation set, this means it will automatically adjust to domain shifts and maintain predictive quality over time.
- ☐ The distribution of the patient features x with distribution $p(x)$ is never affected by domain shifts.
- ☐ During disease waves of infectious diseases (such as COVID-19), the prevalence of a disease and thus the probability of observing this phenotype $p(y)$ changes.
- ☐ Assume that the diagnostic process to determine the disease status y is changed, e.g. an antigen test is replaced by a PCR test. This represents a concept shift $p(y | x)$.

Answers

- ✗ **A.** A good *random* validation score says nothing about future shifts; models do *not* auto-adapt over time.
- ✗ **B.** “Never” is wrong — $p(x)$ *does* shift (covariate shift), e.g. a different patient population.
- ✓ **C.** A disease wave changes the prevalence $p(y)$ — a prior/label shift.
- ✓ **D.** Redefining how the label is measured changes the $x \rightarrow y$ relationship: a concept shift $p(y | x)$.

Question 27: Medical imaging modalities

Which statements about medical imaging modalities are correct?

- ☐ Computed Tomography (CT) scans use ultrasound images to generate images of the body.
- ☐ X-rays are used to differentiate between tissues of varying densities, such as bone and soft tissue.
- ☐ Magnetic Resonance Imaging (MRI) provides detailed images of internal structures by using strong magnetic fields and radio waves.
- ☐ Optical coherence tomography imaging uses multiple X-rays to create images of organs and tissues.

Answers

- ✗ **A.** CT is built from many *X-ray* projections, not ultrasound.
- ✓ **B.** X-ray contrast comes from differential absorption by tissue density (bone vs. soft tissue).
- ✓ **C.** MRI uses strong magnetic fields + radio-frequency pulses (no ionising radiation).
- ✗ **D.** OCT uses *light* (near-infrared interferometry), not X-rays.

Question 28: Most common task in medical imaging

Compared with typical computer-vision tasks, which task is especially prominent in medical imaging?

- ☐ Image generation.
- ☐ Classification.
- ☐ Segmentation.
- ☐ Object detection.

Answers

- ✗ **A.** Image generation is a niche/auxiliary use, not the dominant clinical task.
- ✓ **B.** Classification (e.g. disease present/absent, grading) is also very common.
- ✓ **C.** *Segmentation* (delineating organs/lesions pixel-by-pixel) is the signature, disproportionately common medical-imaging task — hence the U-Net.
- ✗ **D.** Object detection is comparatively rare in medical imaging versus segmentation and classification.

Question 29: The U-Net architecture

Which of the following correctly describe the U-Net architecture?

- ☐ The U-Net was the first vision model where the encoder is based on self-attention instead of convolutions.
- ☐ The U-Net architecture uses a symmetrical encoder and decoder.
- ☐ The U-Net has been specifically developed for the object detection task.

- ☐ To aid the decoding process, feature maps from the encoder are concatenated with feature maps of the corresponding up-sampling level of the decoder.

Answers

- ✗ **A.** U-Net is fully *convolutional*; it predates and does not use self-attention encoders.
- ✓ **B.** The “U” is a symmetric contracting encoder + expanding decoder.
- ✗ **C.** It was designed for *segmentation* (biomedical), not object detection.
- ✓ **D.** Skip connections concatenate encoder feature maps into the matching decoder level — the key idea that preserves spatial detail.

Question 30: The “Clever Hans” effect

In the context of AI for medical imaging, what does the “Clever Hans” effect refer to?

- ☐ A bias occurring when enhancing the resolution of low-quality medical images.
- ☐ The situation where an AI model appears to perform well but is actually relying on irrelevant features or artifacts in the data.
- ☐ The phenomenon where an AI system incorrectly predicts outcomes based on biased training data.
- ☐ The phenomenon where data augmentation is used to improve the robustness of AI models in medical imaging.

Answers

- ✗ **A.** Unrelated to super-resolution bias.
- ✓ **B.** Exactly shortcut learning: the model *looks* accurate but is exploiting an irrelevant artifact (e.g. a hospital tag/ruler), not the pathology.
- ✗ **C.** Tempting, but too generic: “biased training data” describes *any* bias. Clever Hans specifically means relying on *spurious shortcut features* while appearing to perform well.
- ✗ **D.** Data augmentation is a mitigation technique, not the Clever Hans effect.

Question 31: Explainable AI and interpretability in medical imaging

Which statements about explainable AI and interpretability in medical imaging are correct?

- ☐ Gradient integration utilizes the propagation of a relevance value back to the input layers, layer by layer, to estimate the relevance of each input feature.
- ☐ Explainable AI techniques, such as Grad-CAM, help visualize which parts of a medical image are most important for a model’s prediction.
- ☐ Occlusion analysis involves removing patches of the training data images to improve the model’s interpretability.
- ☐ Counterfactual explanations in medical imaging involve generating alternative images to show what changes would have been needed for a different diagnosis.

Answers

- ✗ **A.** The layer-by-layer back-propagation of a *relevance* value is Layer-wise Relevance Propagation (LRP); “gradient integration” is the wrong name for it.
- ✓ **B.** Grad-CAM produces a heatmap of the image regions most responsible for the prediction.
- ✗ **C.** Occlusion perturbs the *input* image at inference (mask a patch, see the score drop) — it does *not* remove training data, and it explains rather than “improves interpretability”.
- ✓ **D.** Counterfactuals show the minimal image change that would flip the diagnosis — a what-if explanation.

Question 32: AlphaFold 2

Which of the following statements about AlphaFold 2 are correct?

- ☐ AlphaFold 2 incorporates quantum mechanical calculations to improve the accuracy of its predictions.
- ☐ AlphaFold 2 incorporates multiple sequence alignments and homologous protein structures to improve the accuracy of its predictions.
- ☐ AlphaFold 2 is a deep learning model designed to predict the three-dimensional structure of proteins from their amino acid sequences.
- ☐ AlphaFold 2 uses a convolutional neural network architecture.

Answers

- ✗ **A.** No quantum-mechanical calculations are used — it is a learned model.
- ✓ **B.** It relies heavily on multiple sequence alignments (co-evolution signal) and template structures.
- ✓ **C.** Its job is exactly sequence → 3D structure prediction.
- ✗ **D.** The architecture is *attention*-based (the Evoformer + structure module), not a CNN.

Biological Sequences**Question 33: Composition of biological sequences**

Which statements about the composition of biological sequences are correct?

- ☐ Protein sequences are composed of chains of amino acids linked by peptide bonds.
- ☐ DNA sequences are composed of nucleic acids.
- ☐ Biological sequences do not follow any specific patterns or rules in their composition.
- ☐ RNA sequences are composed of lipids.

Answers

- ✓ **A.** Proteins are amino-acid chains joined by peptide bonds.
- ✓ **B.** DNA is a chain of nucleotides (nucleic acid).

- ✗ **C.** Sequences are highly structured (motifs, codons, grammar) — “no rules” is false and is what makes them learnable.
- ✗ **D.** RNA is a *nucleic acid*, not a lipid.

Question 34: Deep learning for biological sequences

Which of the following statements about deep learning methods for biological sequences are correct?

- ☐ Feature extraction from biological sequences can be used to identify motifs, which are recurring patterns or elements within the sequences.
- ☐ One-hot encoding is commonly used to prepare biological sequences as input for deep learning models.
- ☐ Deep learning techniques can be applied to tasks such as classification, segmentation, and generative modeling of biological sequences.
- ☐ 1D-Convolutions, Recurrent Neural Networks (RNNs), Long Short-Term Memory networks (LSTMs), and Transformers are all used to analyze biological sequences.

Answers

- ✓ **A.** Learned filters detect recurring *motifs* (e.g. binding sites).
- ✓ **B.** One-hot encoding of the alphabet (A/C/G/T or 20 amino acids) is the standard input format.
- ✓ **C.** Classification, segmentation (per-position labels) and generation are all applied to sequences.
- ✓ **D.** 1D-CNNs, RNNs, LSTMs and Transformers are all standard sequence architectures here — every option is correct.

2026 exam — single-best-answer

The remaining sections are from the **2026** exam (25 June 2026). Each question has **exactly one** correct option among the three listed; the answer box marks it ✓ and explains why the other two are wrong.

Molecules, Representations & Cheminformatics (2026)

Question 35: Molecular graphs vs. fingerprints

Which of the following statements accurately describes the relationship between molecular graphs and molecular fingerprints?

- ☐ A fingerprint can be derived from a molecular graph, but the exact graph generally cannot be reconstructed from the fingerprint.
- ☐ Fingerprints and molecular graphs necessarily contain the same structural information.
- ☐ A molecular graph can always be reconstructed exactly from a fingerprint.

Answers

- ✓ **A.** A fingerprint is *computed from* the graph (substructure enumeration/hashing), but the mapping is lossy and many-to-one, so the exact graph cannot be recovered.
- ✗ **B.** A fingerprint is a compressed bit/count summary that discards information, so it does *not* carry the same structural information as the graph.
- ✗ **C.** Exact reconstruction is impossible in general — different molecules can even hash to the same bits.

Question 36: Converting between SMILES and fingerprints

Which statement about converting between SMILES strings and molecular fingerprints is correct?

- ☐ Both SMILES and molecular fingerprints can be converted into each other using cheminformatics tools.
- ☐ Only a SMILES can be converted into a molecular fingerprint using cheminformatics tools.
- ☐ Only a molecular fingerprint can be converted into a SMILES using cheminformatics tools.

Answers

- ✗ **A.** The round-trip does not exist: you cannot go fingerprint → SMILES.
- ✓ **B.** SMILES → molecular graph → fingerprint is a standard cheminformatics pipeline.
- ✗ **C.** Backwards — fingerprints are lossy, so fingerprint → SMILES is not possible.

Question 37: Tanimoto coefficient

How does the Tanimoto coefficient quantify the similarity of two molecular fingerprints?

- ☐ It divides the number of shared features by the total number of features present in either molecule.
- ☐ It aligns three-dimensional pharmacophore points and calculates their average spatial distance.
- ☐ It measures the overlap between the core scaffolds and assigns a score of one whenever the scaffolds match.

Answers

- ✓ **A.** Tanimoto = $|A \cap B| / |A \cup B|$: shared on-bits divided by the union of on-bits.
- ✗ **B.** Averaging 3D pharmacophore-point distances is a different (3D alignment) similarity, not Tanimoto.
- ✗ **C.** Tanimoto does not test scaffold identity; it counts shared fingerprint features.

Question 38: Aspirin SMILES

Which of the following SMILES strings represents aspirin (acetylsalicylic acid)?

- ☐ O=C(C)Oc1ccccc1C(=O)O

☐ cc(=O)Oc1ccccc1c(=O)O

☐ O=C[C]Oc1ccccc1C[=O]O

Answers

- ✓ **A.** O=C(C)Oc1ccccc1C(=O)O: an acetyl ester (O=C(C)O) on an aromatic ring (c1ccccc1) bearing a carboxylic acid (C(=O)O) — acetylsalicylic acid.
- ✗ **B.** Lowercase *c* marks *aromatic* atoms, so writing the carbonyl carbons as aromatic (cc(=O)) is chemically and syntactically wrong.
- ✗ **C.** [C] and [=O] are malformed SMILES atom tokens — the string does not parse.

Question 39: Bioactivity datasets

Which description of typical bioactivity datasets is the most accurate?

- ☐ They are generally large, balanced, and nearly free of measurement noise.
- ☐ They are often limited, imbalanced, and noisy, with many more inactive than active compounds.
- ☐ They are inexpensive to generate and can therefore cover most combinations of compounds and assays.

Answers

- ✗ **A.** The opposite of reality — bioactivity data is rarely large, balanced, or clean.
- ✓ **B.** It is typically limited, imbalanced (far more inactives than actives), and noisy.
- ✗ **C.** Assays are expensive, so coverage of compound×assay combinations is very sparse.

Question 40: The Design–Make–Test–Analyse cycle

Which statement correctly describes the Design–Make–Test–Analyse (DMTA) cycle in drug discovery?

- ☐ Compounds are tested before they are designed, and only active compounds are subsequently synthesized.
- ☐ Compounds are designed and synthesized once, after which testing replaces any need for further design.
- ☐ Compounds are designed, synthesized, experimentally tested, and the results are analysed to guide the next design cycle.

Answers

- ✗ **A.** Designing comes before testing — you cannot test a compound you have not yet designed and made.
- ✗ **B.** It is an iterative *cycle*; one round of testing does not remove the need for further design.
- ✓ **C.** Design → Make → Test → Analyse, and the analysis feeds the next design iteration.

Virtual Screening, QSAR & Targets (2026)

Question 41: Structure-based vs. ligand-based drug design

Which statement correctly distinguishes structure-based drug design (SBDD) from ligand-based drug design (LBDD)?

- ☐ SBDD requires known active ligands, whereas LBDD requires the 3D structure of the target.
- ☐ SBDD and LBDD both require a high-resolution 3D structure of the target.
- ☐ SBDD uses the 3D structure of the target, whereas LBDD can use information from known ligands without requiring that target structure.

Answers

- ✗ A. Swapped: SBDD needs the target structure, LBDD needs known ligands.
- ✗ B. LBDD specifically does *not* require the target's 3D structure.
- ✓ C. SBDD uses the target's 3D structure; LBDD relies on known ligands without it.

Question 42: Similarity search vs. model-based search

Which statement correctly distinguishes similarity search from model-based search?

- ☐ Similarity search is ligand-based, whereas model-based search necessarily requires the three-dimensional structure of the target binding site.
- ☐ Similarity search retrieves compounds similar to a query molecule, whereas model-based search uses a supervised model learned from labelled examples to predict activity.
- ☐ Similarity search learns a decision function from labelled active and inactive compounds, whereas model-based search ranks compounds by proximity to a query molecule.

Answers

- ✗ A. Model-based search does *not* necessarily need the target's 3D structure — it can be a ligand-based supervised model.
- ✓ B. Similarity search retrieves neighbours of a query molecule; model-based search predicts activity with a model trained on labels.
- ✗ C. The two definitions are reversed.

Question 43: Early-recognition metrics (odd one out)

Which metric is *not* specifically designed to emphasise early recognition of active compounds near the top of a ranked virtual-screening list?

- ☐ BEDROC.
- ☐ AUC-ROC.
- ☐ Enrichment factor.

Answers

- ✗ **A.** BEDROC is built specifically to reward early recognition.
- ✓ **B.** AUC-ROC weights all ranking positions equally, so it is *not* an early-recognition metric — the odd one out.
- ✗ **C.** The enrichment factor measures actives in the top $x\%$ — an early-recognition metric.

Question 44: Evaluating a virtual-screening model

A virtual-screening model ranks one million compounds, but only the top 1% can be tested experimentally. Which evaluation strategy is most appropriate for selecting the model?

- ☐ Select the model with the highest accuracy after applying a 0.5 probability threshold, because ranking quality above that threshold is irrelevant once compounds are classified.
- ☐ Select the model with the highest global AUC-ROC, because equal weighting of all ranking positions guarantees the best enrichment within the top 1%.
- ☐ Prioritize an early-recognition metric such as enrichment factor near the 1% cutoff or BEDROC, because performance at the top of the ranking determines experimental utility.

Answers

- ✗ **A.** A 0.5-threshold accuracy ignores how well the very top of the ranking is ordered — which is all that matters here.
- ✗ **B.** Global AUC-ROC does *not* guarantee good enrichment inside the top 1%.
- ✓ **C.** Use an early-recognition metric — enrichment factor near the 1% cutoff, or BEDROC.

Question 45: Defensible generalization estimates in QSAR

Which workflow gives the most defensible estimate of generalization performance in a QSAR pipeline?

- ☐ Define the data split first, perform feature and hyperparameter selection using training data, and use the held-out test set only for final evaluation.
- ☐ Split the data first, compare model and hyperparameter choices on the test set, and report the best test-set result.
- ☐ Select features on the complete dataset, then split the reduced data and tune hyperparameters using only the training subset.

Answers

- ✓ **A.** Split first, do all feature/hyperparameter selection on the training data, and touch the test set only once for the final estimate.
- ✗ **B.** Tuning on the test set leaks it into model selection — the reported number is optimistic.
- ✗ **C.** Selecting features on the full dataset before splitting also leaks test information.

Question 46: Compound-series bias

What is compound-series bias, and why is it problematic in QSAR modelling?

- ☐ Many compounds are highly structurally related, so a random split can place close analogues in training and test sets and underestimate generalization error.
- ☐ Information from test compounds is used during feature or hyperparameter selection, so the reported performance is biased even when compounds are structurally unrelated.
- ☐ The active class is much smaller than the inactive class, so performance is dominated by the majority class even when chemical series are diverse.

Answers

- ✓ **A.** Close analogues land in both train and test under a random split, so the test set is partly memorised and generalization error is underestimated.
- ✗ **B.** That describes generic information leakage in feature/hyperparameter selection, not compound-series bias.
- ✗ **C.** That describes class imbalance, a different problem.

Question 47: Drug synergy and model symmetry

Which statement correctly describes drug synergy together with an appropriate property of a synergy-prediction model?

- ☐ The combined effect exceeds the additive effect of the separate compounds, and the prediction should not change when the order of the compounds is swapped.
- ☐ The combined effect equals the additive expectation, and the prediction may change when the same pair is supplied in reverse order.
- ☐ The combined effect is smaller than the additive expectation, and the prediction should be invariant to the order of the compounds.

Answers

- ✓ **A.** Synergy is super-additive (combined > additive), and a sensible model is symmetric — swapping the two compounds must not change the prediction.
- ✗ **B.** “Equals the additive expectation” is *no* interaction, and order-dependence is an undesirable property.
- ✗ **C.** “Smaller than additive” is *antagonism*, not synergy (the symmetry property is right, but the definition is wrong).

Question 48: Proteochemometric modelling

What distinguishes proteochemometric modelling from classical single-target QSAR?

- ☐ It jointly uses representations of molecules and protein targets.
- ☐ It uses molecular representations with a softmax output over candidate protein targets.
- ☐ It uses molecular representations with a separate prediction head for each protein target.

Answers

- ✓ **A.** Proteochemometrics jointly encodes the *molecule and the protein target*, so one model spans many targets.
- ✗ **B.** A softmax over targets from the molecule alone has no target representation — not the proteochemometric idea.
- ✗ **C.** A separate head per target (multi-task) still uses only molecular inputs — not a joint molecule+protein representation.

Question 49: Target deconvolution / target fishing

What is target deconvolution, also known as target fishing?

- ☐ Given a protein and a molecular library, rank all molecules by predicted affinity to that one protein.
- ☐ Given a molecule and multiple candidate proteins, predict which protein the molecule is most likely to bind.
- ☐ Given a molecule, independently predict binding to every candidate protein so that any number of targets may be assigned simultaneously.

Answers

- ✗ **A.** Ranking a whole library against *one* protein is virtual screening — the inverse problem.
- ✓ **B.** Target fishing: given a molecule and several candidate proteins, identify the *most likely* target.
- ✗ **C.** Predicting binding to *every* target independently is multi-task target prediction, not deconvolution's "single most likely target" framing.

Graph Neural Networks (2026)

Question 50: Reordering atoms in a molecular graph

The atoms of a molecular graph are relabelled (reordered) without altering its connectivity. How should a graph neural network behave?

- ☐ Node representations should be reordered in the same way, but the graph-level prediction may change because the adjacency matrix has been reordered.
- ☐ Both node representations and the graph-level prediction should remain in exactly the same tensor positions.
- ☐ Node representations should be reordered in the same way, while a graph-level prediction should remain unchanged.

Answers

- ✗ **A.** A pure relabelling must *not* change the graph-level prediction — it is the same graph, just reindexed.
- ✗ **B.** Node representations should follow the permutation, not stay in fixed positions.

- ✓ **C.** Correct symmetry: node features are *equivariant* (they permute with the atoms) while the graph-level readout is *invariant*.

Question 51: Steps of an MPNN layer

Which option correctly lists the typical steps performed in one Message Passing Neural Network (MPNN) layer?

- ☐ Message generation, aggregation, update.
- ☐ Message generation, graph convolution, aggregation.
- ☐ Graph convolution, message generation, update.

Answers

- ✓ **A.** One MPNN layer: generate messages, aggregate them at each node, then update the node state.
- ✗ **B.** “Graph convolution” is not a separate sub-step here, and aggregation is not the final step.
- ✗ **C.** Wrong order and wrong steps — the update must come last.

Question 52: Oversmoothing in GNNs

After stacking many message-passing layers, the node representations in a molecular graph become almost indistinguishable. Why is this problematic?

- ☐ Long-range information is compressed through narrow graph bottlenecks, so distant signals cannot be transmitted effectively.
- ☐ The model may lose the ability to distinguish local atom environments because node embeddings become too similar.
- ☐ Important information fails to reach distant nodes because too few message-passing steps were used.

Answers

- ✗ **A.** Distant signals failing through bottlenecks is *over-squashing*, a different pathology.
- ✓ **B.** Oversmoothing: after many layers the node embeddings converge, so the model can no longer tell local atom environments apart.
- ✗ **C.** “Too few steps” (under-reaching) is the opposite of the “many layers” premise.

Question 53: Contrastive learning

How does contrastive learning construct a shared molecule–target representation space?

- ☐ It clusters molecules and targets separately and assumes clusters with the same index correspond to binding pairs.
- ☐ It increases similarity for associated molecule–target pairs and decreases similarity for mismatched pairs.
- ☐ It concatenates molecule and target representations and fits a supervised activity regressor without comparing pairs in a shared space.

Answers

- ✗ **A.** Clustering each modality separately and matching by cluster index is not contrastive learning.
- ✓ **B.** Contrastive learning pulls associated molecule–target pairs together and pushes mismatched pairs apart in a shared space.
- ✗ **C.** Concatenating and fitting a supervised regressor never builds a shared comparison space.

Question 54: Multiple Instance Learning

In what way does Multiple Instance Learning differ from standard supervised learning? Instead of feature vectors each labelled with a single value $(\mathbf{x}, y)$, ...

- ☐ ... we have a bag of feature vectors labelled with a single value $(\{\mathbf{x}_1, \dots, \mathbf{x}_N\}, y)$.
- ☐ ... we have a bag of feature vectors labelled with a bag of values $(\{\mathbf{x}_1, \dots, \mathbf{x}_N\}, \{y_1, \dots, y_M\})$.
- ☐ ... we have feature vectors labelled with a bag of values $(\mathbf{x}, \{y_1, \dots, y_M\})$.

Answers

- ✓ **A.** MIL replaces $(\mathbf{x}, y)$ with a *bag* of vectors sharing one label, $(\{\mathbf{x}_1, \dots, \mathbf{x}_N\}, y)$.
- ✗ **B.** A bag labelled with a *bag* of values is not MIL (that would be a multi-output/multi-label setup).
- ✗ **C.** A single vector with multiple labels is ordinary multi-label learning, not MIL.

Generative Models & Synthesis Planning (2026)

Question 55: Reward hacking a QSAR proxy

A generator is optimized against a QSAR-based activity score. It produces molecules with very high predicted activity, but follow-up assays show poor activity. What is the most likely explanation?

- ☐ The generator received the activity score only after completing each molecule, making the reward delayed.
- ☐ The generator exploited weaknesses of the proxy scoring model rather than improving true experimental activity.
- ☐ The generator produced too few structurally distinct molecules, reducing diversity among candidates.

Answers

- ✗ **A.** Delayed (sparse) reward does not explain why *predicted* activity is high while *real* activity is low.
- ✓ **B.** Classic reward hacking: the generator found inputs that fool the proxy QSAR model instead of being genuinely active.
- ✗ **C.** Low diversity would not systematically inflate predicted activity above true activity.

Question 56: Combating analogue rediscovery (odd one out)

A goal-directed generator repeatedly rediscovers close analogues of earlier high-scoring molecules. Which intervention is *not* an approach to combat this behaviour?

- ☐ Mix samples from a fixed exploratory generator with samples from the trainable generator during SMILES generation.
- ☐ Penalize the reward for molecules that are similar to previously generated high-scoring molecules stored in a memory buffer.
- ☐ Increase exploitation by always selecting the molecule with the highest predicted score as the next training example.

Answers

- ✗ **A.** Mixing in a fixed exploratory generator adds diversity — a genuine remedy, so not the answer.
- ✗ **B.** Penalising similarity to a memory of past hits also promotes diversity — a genuine remedy.
- ✓ **C.** This is the answer: always taking the single highest-scoring molecule *increases* exploitation and worsens rediscovery, so it is *not* a remedy.

Question 57: Autoregressive molecular reinforcement learning

In autoregressive molecular generation framed as reinforcement learning, which mapping of RL concepts is correct?

- ☐ The partial molecule is the state and the next construction step is the action, but the scoring function is the policy and the generator supplies the reward.
- ☐ The completed molecule is the state, the scoring function is the action, and each partial molecule receives an independent final reward.
- ☐ The partial molecule is the state, the next construction step is the action, the generator is the policy, and the completed molecule receives a reward.

Answers

- ✗ **A.** The *generator* is the policy, not the scoring function, and the generator does not “supply the reward”.
- ✗ **B.** The completed molecule is not the state, and partial molecules do not each get an independent final reward.
- ✓ **C.** Partial molecule = state, next construction step = action, generator = policy, completed molecule earns the reward.

Question 58: Diffusion vs. one-shot latent decoding

What distinguishes diffusion-based molecular generation from one-shot latent decoding?

- ☐ It maps a single latent vector directly to a complete molecule in one decoder pass.
- ☐ It starts from noise and iteratively applies a learned denoising process.
- ☐ It generates a molecule step by step by predicting the next token or graph action from

the partial molecule.

Answers

- ✗ **A.** Mapping one latent vector to a full molecule in a single pass is exactly one-shot decoding.
- ✓ **B.** Diffusion starts from noise and *iteratively* denoises toward a molecule.
- ✗ **C.** Step-by-step next-token / graph-action generation is autoregressive, not diffusion.

Question 59: Conditional generation

A generator receives a desired logP value as an input and produces molecules in one pass. It was trained on molecule–logP pairs and does not evaluate a logP scoring function during generation. Which paradigm does this describe?

- ☐ Conditional generation.
- ☐ Goal-directed generation.
- ☐ Distribution learning followed by unconditional sampling.

Answers

- ✓ **A.** The property is given as a *condition*, the model maps it to molecules in one pass, and no scoring function is queried — conditional generation.
- ✗ **B.** Goal-directed generation optimises against a scoring function during generation, which is explicitly excluded here.
- ✗ **C.** Unconditional sampling would ignore the desired logP input.

Question 60: Distribution learning

Which statement best describes distribution learning in molecular generation?

- ☐ Distribution learning generates molecules conditioned on auxiliary information such as protein targets or desired property values.
- ☐ Distribution learning aims to generate molecules that resemble those in the training dataset by learning their underlying distribution.
- ☐ Distribution learning focuses on generating molecules that are optimized for specific biological or physicochemical properties.

Answers

- ✗ **A.** Conditioning on auxiliary information is *conditional* generation.
- ✓ **B.** Distribution learning models the training distribution and samples molecules resembling it.
- ✗ **C.** Optimising for specific properties is *goal-directed* generation.

Question 61: Mode collapse

Which statement best describes mode collapse in generative molecule models?

- ☐ The model generates molecules with high validity and novelty but poor agreement with the training-set property distribution.
- ☐ The model repeatedly generates the same or very similar molecules.
- ☐ The model generates almost exclusively molecules that already occur in the training dataset.

Answers

- ✗ **A.** Good validity/novelty but poor distribution match is a different failure mode, not mode collapse.
- ✓ **B.** Mode collapse: the generator keeps producing the same or very similar molecules (it covers only a few “modes”).
- ✗ **C.** Re-emitting training molecules is memorisation, not mode collapse.

Question 62: Reaction prediction and retrosynthesis

Which mapping between task and input/output is correct?

- ☐ Both forward reaction prediction and single-step retrosynthesis take only reaction conditions as input and predict reaction yield.
- ☐ Forward reaction prediction maps reactants to products, while single-step retrosynthesis maps a target product to possible precursors.
- ☐ Forward reaction prediction maps products to reactants, while single-step retrosynthesis predicts the major product from known reactants.

Answers

- ✗ **A.** Neither task takes only conditions and predicts yield.
- ✓ **B.** Forward prediction: reactants → products; single-step retrosynthesis: product → precursors.
- ✗ **C.** The two directions are swapped.

Question 63: Reaction templates

What is the purpose of a reaction template?

- ☐ To provide a generalized symbolic rule describing required substructures and how reactants are transformed into products.
- ☐ To encode reaction conditions such as solvent and temperature, while leaving the structural transformation to a neural model.
- ☐ To assign a reaction class label that groups similar reactions, without specifying how atoms or bonds are transformed.

Answers

- ✓ **A.** A reaction template is a generalised symbolic rule: required substructures plus how reactants become products.
- ✗ **B.** Encoding solvent/temperature conditions is not what a template captures.
- ✗ **C.** Assigning a reaction *class* without the atom/bond transformation is classification, not a template.

Question 64: Easiest reaction task

Which of these tasks is generally the easiest?

- ☐ Single-step retrosynthesis.
- ☐ Multi-step retrosynthesis planning.
- ☐ Forward reaction prediction.

Answers

- ✗ **A.** Single-step retrosynthesis is one-to-many (many valid precursor sets), so it is harder.
- ✗ **B.** Multi-step planning searches over many steps — the hardest of the three.
- ✓ **C.** Forward reaction prediction is the most constrained (reactants → major product) and generally the easiest.

Question 65: Template-free translation methods

How do template-free translation methods frame reaction prediction or retrosynthesis?

- ☐ As retrieval of the most compatible stored reaction pattern from associative memory.
- ☐ As classification of the input into a learned reaction category followed by symbolic transformation.
- ☐ As direct sequence-to-sequence translation between molecular reaction strings.

Answers

- ✗ **A.** Retrieving stored patterns from memory is a template/retrieval approach.
- ✗ **B.** Classifying into a learned reaction class then transforming is template-based.
- ✓ **C.** Template-free methods translate reaction strings directly, as sequence-to-sequence (e.g. the Molecular Transformer).

Question 66: Deep networks inside MCTS for retrosynthesis

Why are deep neural networks useful inside Monte Carlo Tree Search for retrosynthesis?

- ☐ They replace tree search by evaluating only the first retrosynthetic disconnection and then executing that route greedily.
- ☐ They can estimate the value of unvisited states and suggest promising actions when exhaustive search is infeasible.
- ☐ They mainly balance chemical equations at each node, while the search itself decides

value and policy without learned guidance.

Answers

- ✗ **A.** They do not collapse the search to a single greedy first disconnection.
- ✓ **B.** Learned networks estimate the value of unvisited states and propose promising actions, guiding the search where exhaustive enumeration is infeasible.
- ✗ **C.** They provide the value/policy guidance — the search is not left unguided.

Question 67: Pocket-conditioned 3D ligand generation

Which statement best describes pocket-conditioned 3D ligand generation?

- ☐ Ligands are generated without protein context and docked into the pocket only after generation; the docking score does not condition the generator.
- ☐ The protein binding pocket provides geometric context for generating or denoising a ligand, often followed by validity, docking, property, and synthesis checks.
- ☐ A fixed ligand graph is provided and only alternative conformations of that same molecule are sampled without using protein information.

Answers

- ✗ **A.** Generating without protein context and only docking afterwards is *not* pocket-conditioned — the pocket never conditions the generator.
- ✓ **B.** The binding pocket gives geometric context to generate/denoise the ligand, typically followed by validity, docking, property, and synthesis checks.
- ✗ **C.** Sampling conformers of a fixed ligand without protein information is not pocket-conditioned generation.